Basic Linux Commands

Printing Working Directory - pwd

- The pwd (print working directory) command displays a user's current location in the directory tree.

Changing Directories - cd

- The cd (change directory) command is used to navigate the directory tree.
- To change directory to /usr/bin:
  - $ cd /usr/bin
- To go back to the home directory, issue either of the following:
  - $ cd or $ cd ~
- To switch between current and previous directories, issue the cd command with the dash character:
  - $ cd –

To List content of file and directories:

- **ls**
  - **–a** Lists hidden files also. If a file or directory name starts with a dot, it is considered hidden.
  - **–lh** Displays long listing with file sizes in human readable format.
  - **–l** Displays long listing with detailed file information including the file type, permissions, link count, owner, group, size, date and time of last modification, and name of the file.
  - **–ld** Displays long listing of the specified directory, but hides its contents.
  - **–R** Lists contents of the specified directory and all its sub-directories (recursive listing).
  - **–lt** Lists all files sorted by date and time with the newest file first.
  - **–ltr** Lists all files sorted by date and time with the oldest file first.
  -

Listing Currently Logged-In Users

- who
- who am i
- w
- logname

Uptime

- The uptime command shows the system's current time, how long it has been up for, number of users currently logged in, and average number of processes over the past 1, 5, and 15 minutes
- System load averages is the average number of processes that are either in a **runnable** or **uninterruptable** state.
  - A process in a runnable state is either using the CPU or waiting to use the CPU.
  - A process in uninterruptable state is waiting for some I/O access, e.g. waiting for disk.
- The averages are taken over the three time intervals. Load averages are not normalized for the number of CPUs in a system, so a load average of 1 means a single CPU system is loaded all the time while on a 4 CPU system it means it was idle 75% of the time.
- The load average is the average number of processes in a wait state over a period of time.

uname – system information

- The uname command produces basic information about the system. Without any options, this command displays the operating system name only.
- Try running the *uname command with the*
  - *–s (kernel name)*
  - *–n (node name)*
  - *–r (kernel release)*
  - *–v (kernel build date)*
  - –m (hardware name)
  - –p (processor type)
  - –i (hardware platform)
  - –o (OS name)

Absolute and Relative Path names

- **Absolute Path**
  - An absolute path (a.k.a. a full path or a fully qualified path name) points to a file or directory in relation to /.
  - It always starts with the forward slash (/) character.
- **Relative path**
  - Points to a file or directory in relation to your current location in the directory tree.
  - A relative path never begins with a forward slash; rather it always begins in one of the following two ways:
    - ✓ With two periods: This represents a parent directory in relation to your current working directory. A parent directory is one level higher than the current working directory. For example, to go one level up to the parent directory, type:
      - # cd ..
    - ✓ With a sub-directory name: Suppose you are in the / directory and want to go to the sysconfig sub-directory under /etc, you would run the following:
      - # cd etc/sysconfig

Creating, Removing, Copying, Moving files & Directories

Creating a file in Linux

Using cat command:

- cat (Concatenate) command is used to create a file and to display and modify the contents of a file.
- To create a file

 # cat > filename (say ytfile)

Welcome to Yoinsights Technologies.

Ctrl+d (To save the file)

```
[root@yoinsights ~]# cat > ytfile
Welcome to Yoinsights Technologies.
```

To display the content of the file

 # cat filename (say ytfile)

```
[root@yoinsights ~]# cat ytfile
Welcome to Yoinsights Technologies.
[root@yoinsights ~]#
```

To append the data in the already existing file

cat >> <filename>

cat >> ytfile

Ctrl+d (to save the changes)

```
[root@yoinsights ~]# cat >> ytfile
Welcome to Yoinsights Technologies.
Welcome to Linux World
[root@yoinsights ~]#
```

Creating multiple files at same time using touch command

#touch <filename> <filename> <filename>

#touch file1 file2 file3

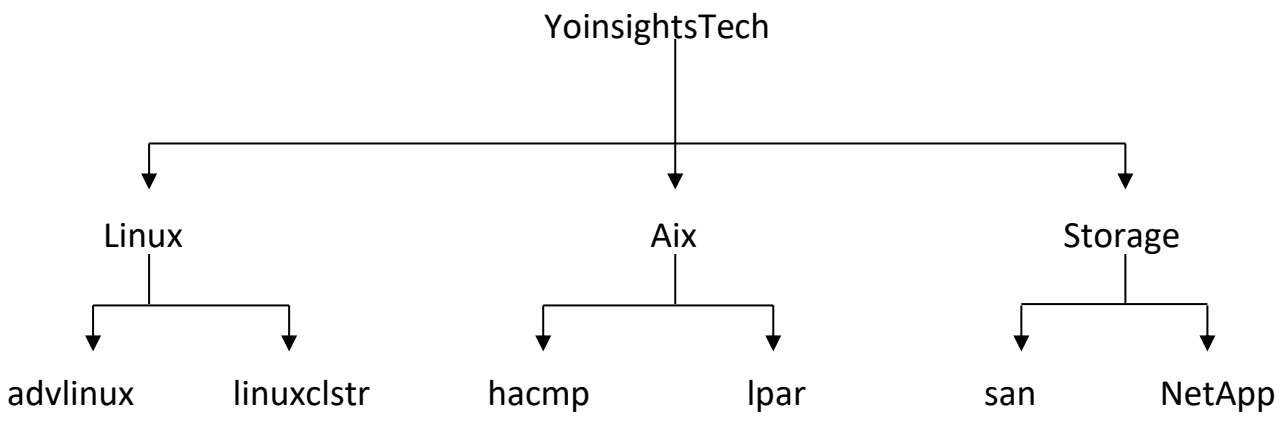
Note: to check the files use

 # ls

```
[root@yoinsights ~]# touch file1 file2 file3
[root@yoinsights ~]# ls
anaconda-ks.cfg  Downloads  file3          Music     Templates
Desktop          file1      install.log    Pictures  Videos
```

```
Documents          file2      install.log.syslog  Public     ytfile
[root@yoinsights ~]#
```

Making multiple directories inside a directory:
Let us make some directories according to the following architecture in one command.



mkdir -p YoinsightsTech/{Linux/{advlinux,linuxclstr},Aix/{hacmp,lpar},Storage/{san,netapp}}

Check it by using **tree** command or **ls –R** command

```
[root@yoinsights ~]# tree
YoinsightsTech/{Linux/{advlinux,linuxclstr},Aix/
{hacmp,lpar},Storage/{san,netapp}} [root@yoinsights ~]# tree
YoinsightsTech/
YoinsightsTech/
├── Aix
│   ├── hacmp
│   └── lpar
├── Linux
│   ├── advlinux
│   └── linuxclstr
└── Storage
    ├── netapp
    └── san
```

Copying files into directory:

#cp <source filename> <destination directory in which to paste the file>

#cp file1 ytdir

```
[root@yoinsights ~]# cp file1 ytdir/
 [root@yoinsights ~]# cd ytdir
```

```
[root@yoinsights ytdir]# ls
file1
```

Create directory with multiple internal files

 # mkdir <dir name>; touch <destination path/file name{number 1..10}>

mkdir ytdir2; touch /root/ytdir2/file {1..5}

```
[root@yoinsights ~]# mkdir ytdir2; touch /root/ytdir2/file{1..5}
[root@yoinsights ~]# tree ytdir2
ytdir2
├── file1
├── file2
├── file3
├── file4
├── file5
```

Copying directories from one location to other

 # cp –rvfp <dir name> <destination name>

#cp –rvfp ytdir2 ytdir

```
[root@yoinsights ~]# cp -rvfp ytdir2 ytdir
`ytdir2' -> `ytdir/ytdir2'
`ytdir2/file4' -> `ytdir/ytdir2/file4'
`ytdir2/file2' -> `ytdir/ytdir2/file2'
`ytdir2/file5' -> `ytdir/ytdir2/file5'
`ytdir2/file1' -> `ytdir/ytdir2/file1'
`ytdir2/file3' -> `ytdir/ytdir2/file3'
[root@yoinsights ~]# cd ytdir
[root@yoinsights ytdir]# ls
file1  ytdir2
```

Moving files from one location to other (cut and Paste):

#mv <Source> <Destination>

#mv file2 ytdir

```
[root@yoinsights ~]# mv file2 ytdir
[root@yoinsights ~]# ls
anaconda-ks.cfg  Downloads  file3                Music     Templates       ytdir
Desktop          file1      install.log          Pictures  Videos          ytdir2
Documents        file10}    install.log.syslog   Public    YoinsightsTech  ytfile
[root@yoinsights ~]# cd ytdir
[root@yoinsights ytdir]# ls
file1  file2
```

Moving a Directory from one location to other:

#mv <Source dir-name> <destination dir-name>

#mv ytdir ytdir2

```
[root@yoinsights ~]# ls
anaconda-ks.cfg  Downloads  file3                Music     Templates      ytdir
Desktop          file1      install.log          Pictures  Videos         ytdir2
Documents        file10}    install.log.syslog   Public    YoinsightsTech ytfile
[root@yoinsights ~]#  mv ytdir ytdir2
[root@yoinsights ~]# ls
anaconda-ks.cfg  Downloads  file3                Music     Templates      ytdir2
Desktop          file1      install.log          Pictures  Videos         ytfile
Documents        file10}    install.log.syslog   Public    YoinsightsTech
[root@yoinsights ~]# cd ytdir2
[root@yoinsights ytdir2]# ls
file1  file2  file3  file4  file5  ytdir
```

Renaming a File :

#mv <old name> <new name>

#mv ytfile yoinsightsfile

```
[root@yoinsights ~]# ls
anaconda-ks.cfg  file1                Music     Videos
Desktop          file3                Pictures  YoinsightsTech
Documents        install.log          Public    ytdir2
Downloads        install.log.syslog   Templates ytfile
[root@yoinsights ~]# cat ytfile
Welcome to Yoinsights Technologies.
[root@yoinsights ~]# mv ytfile yoinsightsfile
```

```
[root@yoinsights ~]# ls
anaconda-ks.cfg  file1                Music     Videos
Desktop          file3                Pictures  yoinsightsfile
Documents        install.log          Public    YoinsightsTech
Downloads        install.log.syslog   Templates ytdir2
[root@yoinsights ~]# cat yoinsightsfile
Welcome to Yoinsights Technologies.
[root@yoinsights ~]#
```

Renaming a Directory:

The procedure and command for renaming the directory is exactly same as renaming a file.

#mv <old name> <new name>

```
#mv ytdir yoinsightsdir
[root@yoinsights ~]# ls
anaconda-ks.cfg  file1              Music      Videos
Desktop          file3              Pictures   yoinsightsfile
Documents        install.log        Public     YoinsightsTech
Downloads        install.log.syslog Templates  ytdir2
[root@yoinsights ~]# mv ytdir2 yoinsightsdir
[root@yoinsights ~]# ls
anaconda-ks.cfg  file1              Music      Videos
Desktop          file3              Pictures   yoinsightsdir
Documents        install.log        Public     yoinsightsfile
Downloads        install.log.syslog Templates  YoinsightsTech
[root@yoinsights ~]#
```

Removing a File:

#rm filename or #rm –f filename (without prompting)

```
[root@yoinsights ~]# ls
anaconda-ks.cfg  file1              Music      Videos
Desktop          file3              Pictures   yoinsightsdir
Documents        install.log        Public     yoinsightsfile
Downloads        install.log.syslog Templates  YoinsightsTech
[root@yoinsights ~]# rm yoinsightsfile
rm: remove regular file `yoinsightsfile'? y
```

Without Prompting:

```
[root@yoinsights ~]# rm -f yoinsightsfile
[root@yoinsights ~]# ls
anaconda-ks.cfg  Downloads  install.log        Pictures   Videos
Desktop          file1      install.log.syslog Public     yoinsightsdir
Documents        file3      Music              Templates  YoinsightsTech
```

Removing an Empty directory:

#rmdir dirname

```
[root@yoinsights ~]# ls
anaconda-ks.cfg  file1              Music      Videos
Desktop          file3              Pictures   yoinsightsdir
Documents        install.log        Public     YoinsightsTech
Downloads        install.log.syslog Templates  ytdir
[root@yoinsights ~]# rmdir ytdir
[root@yoinsights ~]# ls
anaconda-ks.cfg  file1              Music      Videos
Desktop          file3              Pictures   yoinsightsdir
Documents        install.log        Public     YoinsightsTech
Downloads        install.log.syslog Templates
```

Removing a directory with files or directories inside:

A directory which is having some contents inside it cannot be removed by rmdir command. There are two ways to delete the directory with contents.

Remove the contents inside the directory and then run rmdir command

#rm –rf dirname (where r stands for recursive and f stands for forcefully.)

```
[root@yoinsights ~]# ls
anaconda-ks.cfg  file1              Music      Videos
Desktop          file3              Pictures   yoinsightsdir
Documents        install.log        Public     YoinsightsTech
Downloads        install.log.syslog Templates
 [root@yoinsights ~]# rmdir yoinsightsdir
rmdir: failed to remove `yoinsightsdir': Directory not empty
[root@yoinsights ~]# rm -rf yoinsightsdir/
[root@yoinsights ~]# ls
anaconda-ks.cfg  file1              Music      Videos
Desktop          file3              Pictures   YoinsightsTech
Documents        install.log        Public
Downloads        install.log.syslog Templates
```

VIM EDITOR

VI Visual display editor
VIM Visual display editor improved

This is command mode editor for files. Other editors in Linux are emacs, gedit.
vi editor is most popular
It has 3 modes:

1. Command Mode
2. Insert mode (edit mode)
3. Extended command mode

Note: When you open the vim editor, it will be in the command mode by default.

In the command mode the cursor's can be used as
h / l / k / j to move cursor left / right / up / down

Insert Mode:

i	To begin insert mode at the cursor position
I	To insert at the beginning of line
a	To append to the next word's letter
A	To Append at the end of the line

o	To insert a new line below the cursor position
O	To insert a new line above the cursor position

Command Mode:

gg	To go to the beginning of the
G	To go to end of the page
w	To move the cursor forward, w
b	To move the cursor backward,
nw	To move the cursor forward to
nb	To move the cursor backward t
u	To undo last change (word)
U	To undo the previous changes
Ctrl+R	To redo the changes
yy	To copy a line
nyy	To copy n lines (5yy or 4yy)
p	To paste line below the curso
P	To paste line above the curso
dw	To delete the word letter by
x	To delete the world letter by
dd	To delete entire line
ndd	To delete n no. of lines from
/	To search a word in the file

Extended Mode: (Colon Mode)

Extended Mode is used for save and quit or save without quit using “Esc” Key with “:”

Esc+:w	To Save the changes
Esc+:q	To quit (Without saving)
Esc+:wq	To save and quit
Esc+:w!	To save forcefully
Esc+wq!	To save and quit forcefully
Esc+:x	To save and quit
Esc+:X	To give password to the file and remove password
Esc+:20(n)	To go to line no 20 or n
Esc+: se nu	To set the line numbers to the file
Esc+:se nonu	To Remove the set line numbers

To open multiple files in vim editor

#vim –o file1 file2

To switch between files use Ctrl +w

Listing files and directories:

#ls list the file names

#ls -l long listing of the file

#ls –l filename to see the permissions of a particular file

#ls -al	shows the files in ascending order of modification.
#ls p*	All the files start with p.
#ls ?ample	Files with any first character and has ample
#ls -ld l*	Directory listing only
#ls –ld directory name	to see the permissions of a particular directory
#ls [ae]*	First character of the filename must be a or e.
# ls [!ae]* !	Symbol complements the condition that follows. The characters must not be a or e.
#ls [a-m][c-z][4-9]	list all the files in specific range

Types of Files:

Symbol	Type of File
-	Normal file
d	Directory
l	Link file (shortcut)
b	Block file (Hard disk, Floppy disk)
c	Character file (Keyboard, Mouse)

SYMBOLIC LINK:

There are two types of Links:-

Sr.	Soft Link	Hard link
1	Size of link file is equal to no. of characters in the name of original file	Size of both file is same
2	Can be created across the Partition	Can't be created across the partition
3	Inode no. of source and link file is different	Inode no. of both file is same
4	If original file is deleted, link is broken and data is lost	If original file is deleted then also link will contain data
5	SHORTCUT FILE	BACKUP FILE

Creating a soft link:

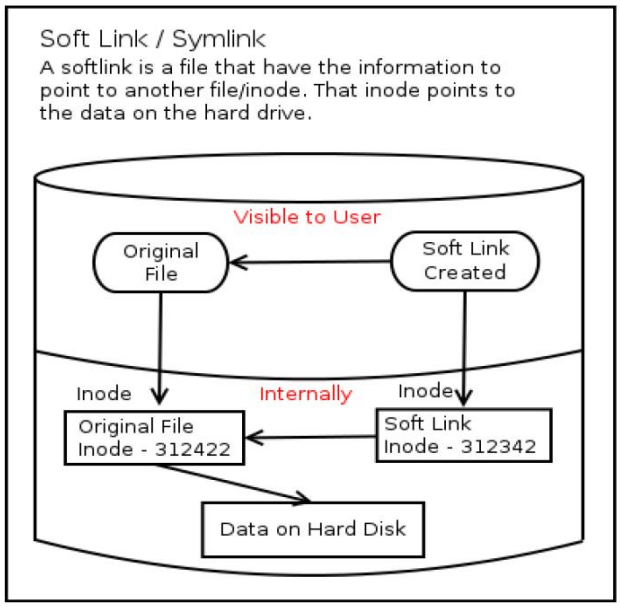
 # ln –s <source file> <destination>

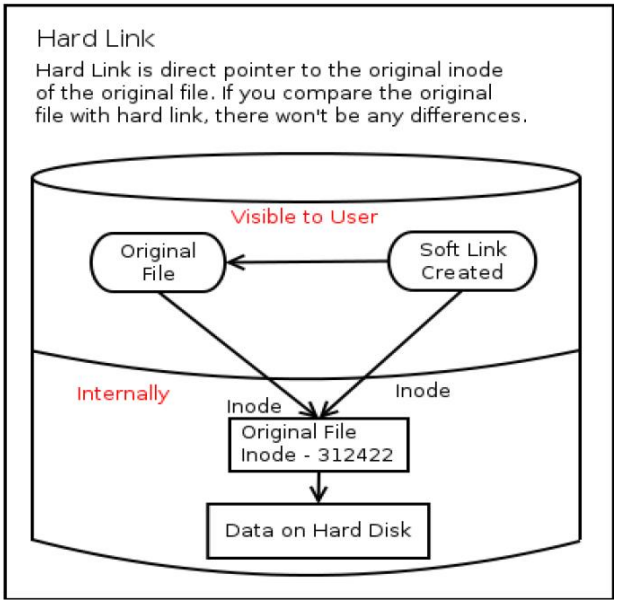
```
[root@yoinsights ~]# ln -s ytfile ytfile.slink
[root@yoinsights ~]# ls -li ytfile ytfile.slink
932158 -rw-r--r--. 1 root root 0 Dec 24 21:51 ytfile
932176 lrwxrwxrwx. 1 root root 6 Dec 24 21:52 ytfile.slink -> ytfile
```

Creating a Hard link:

#ln <source file> <Destination>

```
[root@yoinsights ~]# ln ytfile ytfile.hlink
[root@yoinsights ~]# ls -li ytfile ytfile.hlink
932158 -rw-r--r--. 2 root root 0 Dec 24 22:06 ytfile
932158 -rw-r--r--. 2 root root 0 Dec 24 22:06 ytfile.hlink
```





REGULAR EXPRESSIONS, PIPELINES & I/O REDIRECTIONS

Grep:

Grep stands for **Global Regular Expression Print.** It is used to pick out the required expression from the file and print the output. If grep is combined with another command it can be used to pick out the selected word, phrase from the output of first command and print it.

Examples of Grep:

Let us pick the information about **root** from the file **/etc/passwd** (/etc/passwd contains information about all the users present in the system)

#grep root /etc/passwd

```
[root@yoinsights ~]# grep root /etc/passwd
root:x:0:0:root:/root:/bin/bash
operator:x:11:0:operator:/root:/sbin/nologin
```

To avoid case sensitivity of the word (i.e. the word may be uppercase of lowercase) use **-i**
#grep –i yoinsights ytfile (lets grep the word **yoinsights** whether upper of lower case in the file **ytfile**)

```
[root@yoinsights ~]# grep -i yoinsights ytfile
Welcome to Yoinsights Tech
Welcome to yoinsights Tech
Welcome to YOINSIGHTS TECH
[root@yoinsights ~]#
```

To display a word and 2 lines after the word:

#grep –nA2 wheel /etc/group

```
[root@yoinsights ~]# grep -nA2 wheel /etc/group
11:wheel:x:10:
12-mail:x:12:mail,postfix
13-uucp:x:14:
[root@yoinsights ~]#
```

To display a word and 2 lines after the word:

#grep -nB2 wheel /etc/group

```
[root@yoinsights ~]# grep -nB2 wheel /etc/group
9-mem:x:8:
10-kmem:x:9:
11:wheel:x:10:
```

To display the things except the given word:

#grep –v Yoinsights **ytfile**

```
[root@yoinsights ~]# cat ytfile
Welcome to Yoinsights Tech
Linux is Freedom for everyone
[root@yoinsights ~]# grep -v Yoinsights ytfile
Linux is Freedom for everyone
[root@yoinsights ~]#
```

To display the searched word in color

```
#grep --color root /etc/passwd
```

Combining grep with other commands

```
# cat ytfile | grep –i kernel (Note: pipe | is used to combine to commands)
#ls –l | grep –i ytfile
# ifconfig | grep –i eth0
```

FILTER COMMANDS

- Filter commands are used to filter the output so that the required things can easily be picked up. The commands which are used to filter the output are
  - **#less**
  - **#more**
  - **#head**
  - **#tail**
  - **#sort**
  - **#cut**
  - **#sed**

less:-

The **less** command is used to see the output line wise or page wise.

Ex: less /etc/passwd

```
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
uucp:x:10:14:uucp:/var/spool/uucp:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
gopher:x:13:30:gopher:/var/gopher:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
```

Note: -press **Enter** key to scroll down line by line (or)

Use **d** to go to next page

Use **b** to go to previous page

Use **/** to search for a word in the file

Use **v** to go vi mode where you can edit the file and once you save it you will back to less command

more:-

more is exactly same like **less**

Ex: #more /etc/passwd

Note: -press **Enter** key to scroll down line by line (or)

Use **d** to go to next page

Use **/** to search for a word in the file

Use **v** to go vi mode where you can edit the file and once you save it you will back to more command

head:

It is used to display the top **10 lines** of the file.

Ex: # head /etc/passwd

```
[root@yoinsights ~]# head /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
uucp:x:10:14:uucp:/var/spool/uucp:/sbin/nologin
[root@yoinsights ~]#
```

To display the custom lines

#head -n /etc/passwd (where n can be any number)

```
[root@yoinsights ~]# head -5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
[root@yoinsights ~]#
```

tail:

It is used to display the **last 10** lines of the file

#tail /etc/passwd

```
[root@yoinsights ~]# tail /etc/passwd
haldaemon:x:68:68:HAL daemon:/:/sbin/nologin
gdm:x:42:42::/var/lib/gdm:/sbin/nologin
ntp:x:38:38::/etc/ntp:/sbin/nologin
….
….
tcpdump:x:72:72::/:/sbin/nologin
yogesh:x:500:500::/home/yogesh:/bin/bash
[root@yoinsights ~]#
```

To display the custom lines

#tail -n /etc/passwd (where n can be any number)

```
[root@yoinsights ~]# tail -3 /etc/passwd
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
yogesh:x:500:500::/home/yogesh:/bin/bash
[root@yoinsights ~]#
```

Sort:

It is used to sort the output in numeric or alphabetic order

#sort filename

```
[root@yoinsights ~]# cat ytfile
Welcome to yoinsights Technologies
Welcome to yoinsights Technologies
Welcome to yoinsights Technologies
Linux is Best OS
Linux is Best OS
Linux is Best OS
[root@yoinsights ~]# sort ytfile
Linux is Best OS
Linux is Best OS
Linux is Best OS
Welcome to yoinsights Technologies
Welcome to yoinsights Technologies
Welcome to yoinsights Technologies
[root@yoinsights ~]#
```

To sort the file according to numbers

#sort –d ytfile or #sort –h ytfile

```
[root@yoinsights ~]# cat ytfile
2. Welcome to yoinsights Technologies
3. Welcome to yoinsights Technologies
1. Linux is Best OS
4. Linux is Best OS
[root@yoinsights ~]# sort ytfile
1. Linux is Best OS
2. Welcome to yoinsights Technologies
3. Welcome to yoinsights Technologies
4. Linux is Best OS
 [root@yoinsights ~]#
```

To remove the duplicate entries from the output

#sort –u ytfile

```
[root@yoinsights ~]# cat ytfile
Welcome to yoinsights Technologies
Welcome to yoinsights Technologies
Linux is Best OS
Linux is Best OS
[root@yoinsights ~]# sort -u ytfile
Linux is Best OS
Welcome to yoinsights Technologies
[root@yoinsights ~]#
```

Note:- use **–r** for sorting in reverse order.

cut command:

The cut command is used to pick the given expression (in columns) and display the output.

 # cut -d -f filename (where d stands for delimiter eg. : , “ “ etc… and f stands for field)

```
[root@yoinsights ~]# cut -d: -f1 /etc/passwd
root
bin
daemon
adm
……
……
tcpdump
yogesh
[root@yoinsights ~]# cut -d ':' -f 1,7 /etc/passwd --→ use for multi column
print
```

To delimit spaces and print the field

#cut –d “ ” –f1 filename

To delimit commas and print the field

#cut –d, -f1 filename

To modify the output delimiter use the --output-delimiter=' ' option

cut -d ':' -f 1,7 --output-delimiter=' user having default shell ' /etc/passwd

To check the Architecture of the system

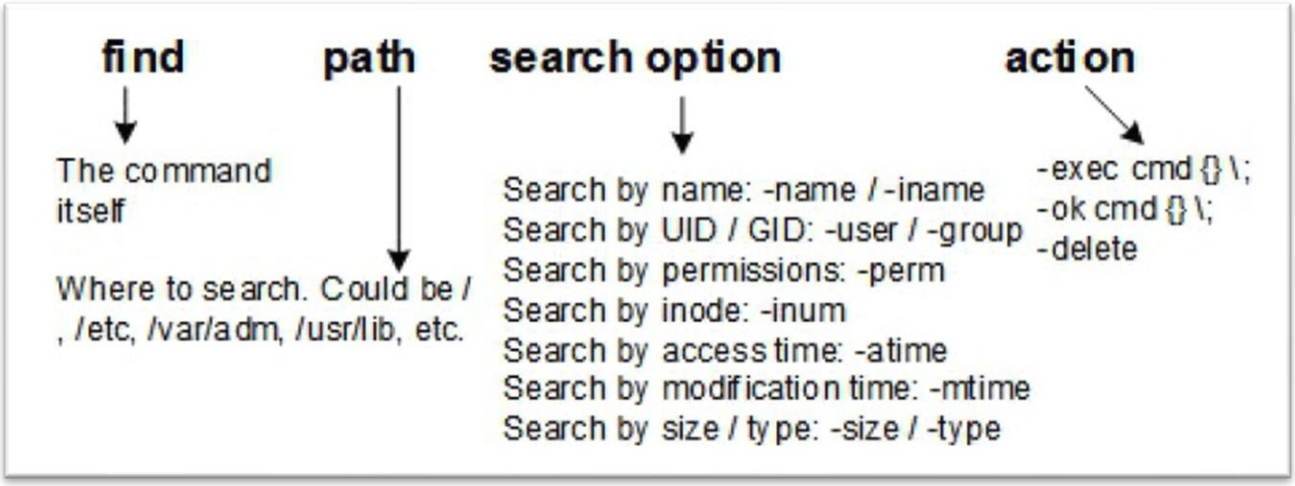
#getconf LONG_BIT

Find command:

The **find** command recursively searches the directory tree, finds files that match the specified criteria, and optionally performs an action on the files as they are found.

Syntax: find / (under root) –option filename

Options that can be used with find command:



Example of find Commands:-

To find files smaller than 1MB in size in root's home directory:

 # find ~ –size –1M

To find files larger than 40MB in size in /usr directory:

 # find /usr –size +40M

To find files in the entire root file system with ownership set to user1 and group membership set to any group other than user1:

 # find / –user user1 –not –group user1

To find files in the /etc directory that were modified more than 2000 days ago:

 # find /etc –mtime +2000

To search for character device files in the /dev directory with permissions set to 666:

 # find /dev –type c –perm 666

To search for core files in the entire directory tree and delete them as they are found without prompting for confirmation:

 # find / –name core –exec rm {} \;

Display Device Information

lsusb – lsusb is a utility for displaying information about USB buses in the system and the devices connected to them

lspci - lspci is a utility for displaying information about PCI buses in the system and devices connected to them.

lscpu - display information about the CPU architecture

dmidecode – shows DMI table decoder

Command Related to Firewall

```
systemctl status firewalld  ---To check the status of firewall
systemctl stop firewalld ---To stop the firewall
systemctl start firewalld ---To start  firewall
systemctl reload firewalld --- reload  firewall
systemctl enable firewalld ---To enable firewall at boot time
systemctl disable firewalld ---To disable firewall
```